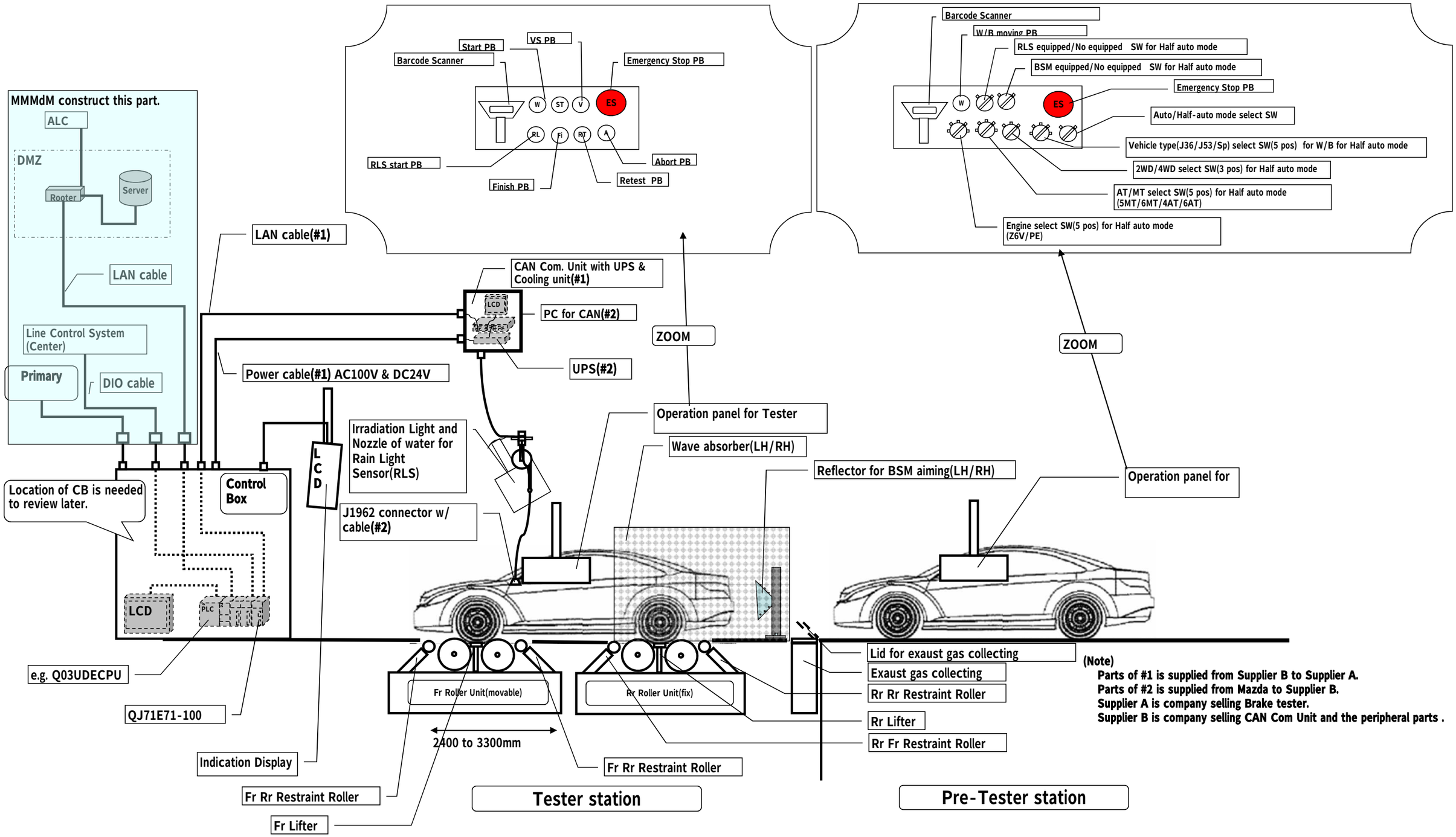
Appendix A1:Layout for Drum tester(Draft)



(Note)
Parts of #1 is supplied from Supplier B to Supplier A.
Parts of #2 is supplied from Mazda to Supplier B.
Supplier A is company selling Brake tester.
Supplier B is company selling CAN Com Unit and the peripheral parts .

Appendix A2:Ethernet between CAN Comunication unit and PLC of Drum Tester

Output From CAN Com unit to PLC

I/O	output signal to PLC	Logic	Condition	Note
M*000	Coupler connection signal(ｶﾌﾟﾗ接続信号)	P 正	Output in case 12V of OBD2 connector is applied to +B terminal of equip side	
M*001	IGN signal(IGN信号)	P 正	Output in case CAN com unit get tester presense positive signal("$7E") from ABS unit.	
M*002	Process OK check signal(処理正常信号)	P 正		
M*003	Process error check signal (処理異常信号)	P 正		
M*004				
M*005	Retest PB confirmation signal(再検査釦確認信号)			
M*006	Abort PB confirmation signal(中止釦確認信号)			
M*007	Reset confirmation signal(ﾘｾｯﾄ確認)			
M*008	FC occurrence check signal(FC発生信号)	P 正	Output in case CAN com unit get current DTC from ABS unit.	
M*009	I/F error signal(I/F異常信号)	N 負	Output In case of CAN com unit is Power-off	
M*010	BSM aiming test requirement confirmation signal		Inspection confirmation	
M*011	"BSM aiming test OK" Display requirement signal		Inspection Requirement	
M*012	"BSM aiming test NG" Display requirement signal		Inspection Requirement	
M*013				
M*014				
M*015	I/F preparation OK signal(I/F準備OK信号)	N	Output in case Application of CAN com unit is runnning normally.This output must blink every 1 sec..	
M*016				
M*017				
M*018				
M*019				
M*020				
M*021				
M*022				
M*023				
M*024				
M*025				
M*026				
M*027				
M*028				
M*029				
M*030				
M*031				

for BSM aiming inspection

Output From PLC to CAN Com unit

I/O	output signal to PLC	Logic	Condition	Note
M*050	Coupler connection confirmation signal(ｶﾌﾟﾗ接続確認信号)	P 正		
M*051	IGN confirmation signal(IGN確認信号)	P 正		
M*052	Process OK check confirmation signal(処理正常確認信号)	P 正		
M*053	Process error check confirmation signal (処理異常確認信号)	P 正		
M*054				
M*055	Retest PB signal(再検査釦信号)			
M*056	Abort PB signal(中止釦信号)			
M*057	Reset signal (ﾘｾｯﾄ信号)			
M*058	FC occurrence check confirmation signal(FC発生確認信号)	P 正		
M*059	I/F error confirmation signal(I/F異常確認信号)	N 負		
M*060	BSM aiming test requirement signal		Inspection requirement	
M*061	"BSM aiming test OK" Display requirement confirmation signal		Inspection confirmation	
M*062	"BSM aiming test NG" Display requirement confirmation signal		Inspection confirmation	
M*063				
M*064				
M*065	DRT running preparation OK signal(BT側運転準備OK信号)	P 正	Output in case Drum tester is runnning normally.This output must blink every 1 sec..	
M*066				
M*067				
M*068				
M*069				
M*070				
M*071				
M*072				
M*073				
M*074				
M*075				
M*076				
M*077				
M*078				
M*079				
M*080				
M*081				

for BSM aiming inspection

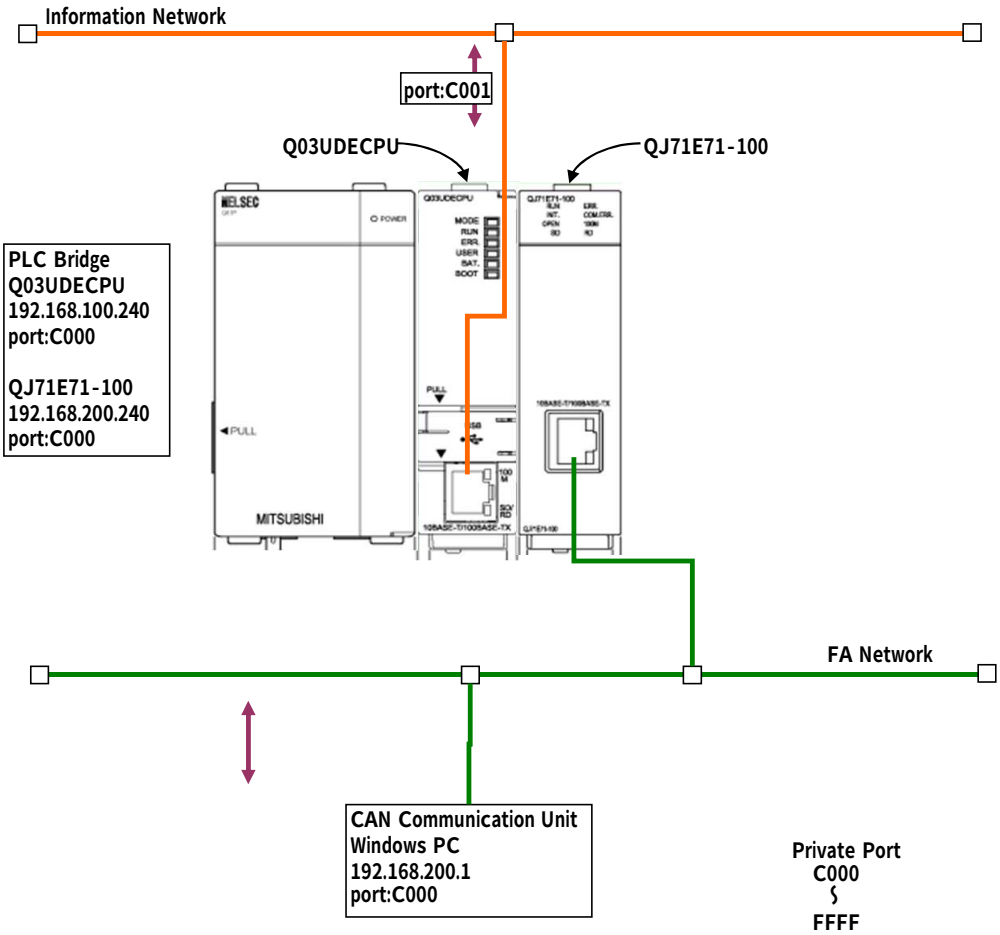
Appendix A3: Exchange information between CAN Comunication unit and PLC of Drum Tester used by Ethernet

Output From CAN Com unit to PLC when test end

(Note) Every data must be sent ASCIIcode.

Device	name of output data to PLC	unit	necessarty numbers of digit	Actual data(example) # is "Space"	CAN/ PLC	The contents of data in case of "**Auto mode**" PLC side need to send these data to ALC system Also, PLC side need to keep record in its own PLC about 500 cars.	The contents of data in case of "**Half-auto mode**" PLC side DOESN'T need to send these dasa to ALC system. But, PLC side need to keep record in its own PLC about 500 cars.	Note
D*000	BSM aiming Test All Result	none	2	"OK" or "NG" or "NA"	CAN side	Set "OK" in case BSM aiming test is all OK. If BSM is NOT equipped, set "NA". Otherwise set "NG"	same as left	
					PLC side	Set this value to 101th-102th of Record-keep format for Inspection results at Drum Tester(Appecdix IT4)	"	
D*001	BSML aiming Test Result	none	2	"OK" or "NG" or "NA"	CAN side	Set "OK" in case BSML aiming test is all OK. If BSML is NOT equipped, set "NA". Otherwise set "NG"	same as left	
					PLC side	Set this value to 103th-104th of Record-keep format for Inspection results at Drum Tester(Appecdix IT4)	"	
D*010 to D*014	DTC code1(BSML)	none	9	"P9024####"	CAN side	In case of existing DTC, set 1st taken place DTC. If there is left digit, set "SP" In case of no DTC, set 'SP'	"	
					PLC side	Set this value to 105th-113th of Record-keep format for Inspection results at Drum Tester((Appecdix IT4)	"	
D*020 to D*024	DTC code2(BSML)	none	9	"#########"	CAN side	In case of existing DTC, set 2nd taken place DTC. If there is left digit, set "SP" In case of no DTC, set 'SP'	"	
					PLC side	Set this value to 114th-122th of Record-keep format for Inspection results at Drum Tester((Appecdix IT4)	"	
D*030	Retry number(BSML)	times	1	"2"	CAN side	Set value of "C1" counter as BSML	"	
					PLC side	Set this value to 123th of Record-keep format for Inspection results at Drum Tester((Appecdix IT4)	"	
D*040 to D*045	Aiming Data(BSML) X1 -Y5	none	12	"0C0C450B3F8C"	CAN side	Set these values of X1,Y1,Y2,Y3,Y4,Y5	"	
					PLC side	Set thiese values to 124th-135th of Record-keep format for Inspection results at Drum Tester((Appecdix IT4)	"	
D*050	BSMR aiming Test Result	none	2	"OK" or "NG" or "NA"	CAN side	Set "NG" in case BSMR aiming test is all OK. If BSMR is NOT equipped, set "NA". Otherwise set "NG"	"	
					PLC side	Set this value to 136th-137th of Record-keep format for Inspection results at Drum Tester(Appecdix IT4)	"	
D*060 to D*064	DTC code1(BSMR)	none	9	"P9024####"	CAN side	In case of existing DTC, set 1st taken place DTC. If there is left digit, set "SP" In case of no DTC, set 'SP'	"	
					PLC side	Set this value to 138th-146th of Record-keep format for Inspection results at Drum Tester((Appecdix IT4)	"	
D*070 to D*074	DTC code2(BSMR)	none	9	"#########"	CAN side	In case of existing DTC, set 2nd taken place DTC. If there is left digit, set "SP" In case of no DTC, set 'SP'	"	
					PLC side	Set this value to 147th-155th of Record-keep format for Inspection results at Drum Tester((Appecdix IT4)	"	
D*080	Retry number(BSMR)	times	1	"2"	CAN side	Set value of "C1" counter as BSMR	"	
					PLC side	Set this value to 156th of Record-keep format for Inspection results at Drum Tester((Appecdix IT4)	"	
D*090 to D*095	Aiming Data(BSMR) X1 -Y5	none	12	"0C0C450B3F8C"	CAN side	Set these values of X1,Y1,Y2,Y3,Y4,Y5	"	
					PLC side	Set thiese values to 124th-135th of Record-keep format for Inspection results at Drum Tester((Appecdix IT4)	"	

Output From PLC to CAN Com unit when test start timing

Device	name of output data to CAN Com unit	unit	necessarty numbers of digit	Actual data	CAN/ PLC	The contents of data from Common Data format from ALC system and PLC Data table in case of selecting "**Auto mode**" of Operation panel;	In case of "**Half-Auto mode**" of Operation panel during abnomal status of ALC system.	Note
D*200 to D*208	VIN	none	17	"$$$$$$$$$$$$$$$$$"	PLC side	Set VIN data from 41th to 57th	None	Reference
D*210 to D*213	AON	none	8	"$$$$$$$"	PLC side	Set VIN data from 9th to 16th	None	Reference
D*220	Car-line	none	2	J36W;TBD	PLC side	Set 66th digit(Vehicle line lower) + 67th digit(Model change level) (note) "Car line" = 66th digit + 67th digit	Set selected positon of "Vehicle type select SW" If selected J36, set value of "Car line" for J36	for near fyture(Additional project etc.)
D*230	Engine	none	2	Z6V;"Z6", PE;"PE"	PLC side	Set 79th digit(Engine 4th-digit) + 80th digit(Engine 3rd-digit)	Set selected positon of "Engine select SW" If selected Z6, set "Z6".	
D*240	2WD/4WD type	none	1	2WD;"X", 4WD; except for "X"	PLC side	Set 190th digit{W20 {4WD equipped?} lower}	Set selected position of "2WD/4WD select SW" If selected 2WD , set value of "X". If selected 4WD, set "TBD".	for near fyture(Additional 4WD etc.)
D*250	AT/MT	none	1	4EAT;"N",6EAT;"A",5MT;"5",6MT;"6"	PLC side	Set 83th digit(Transmission 3rd-digit)	Set selected position of "AT/MT select SW" If selected 5MT , set value of "5".	
D*260	BSM equipped/No equipped	none	1	Equipped; data except for "X",None equipped;"X"	PLC side	Set 194th digit{W22 {BSM equipped?} lower}	Set selected position of "BSM equipped/No equipped select SW" If selected None BSM , set value of "X". If selected equipped BSM, set "TBD".	Must info. for BSM test
					CAN side			If equipped BSM, do BSM aiming test.
D*270	RLS equipped/No equipped	none	1	Equipped; data except for "X",None equipped;"X"	PLC side	Set 158th digit{W04 {RLS parts No} lower}	Set selected position of "RLS equipped/No equipped select SW" If selected None RLS , set value of "X". If selected equipped RLS, set "TBD".	Must info. for BSM test

Appendix A5: Ethernet architecture between PLC and CAN Communication Unit